Mikael Ade Krisna Respati

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LinkedIn | Portfolio | Lampung, Indonesia

Education

Sejong University, Seoul, South Korea *Mar 2024 – Feb 2026*
Master's in AI and Robotics, GPA: 4.14 / 4.50
Thesis: Joint Beamforming and Localization using Multi-Task Deep Learning
Developed a CNN-based model using a soft parameter sharing approach with attention gates and designed an adaptive filter for CSI dimensionality reduction, trained end-to-end with the model. Achieving up to 88% reduction in parameters and 88.01% top-1 beamforming accuracy, 2.30 m localization error.

Gadjah Mada University, Yogyakarta, Indonesia *Aug 2018 – Nov 2022*
Bachelor of Electrical Engineering, GPA: 3.20 / 4.00
Thesis: Development of Indirect Mass Reporting System through Twitter Based on NLP
Developed an SVM-based ML model to classify tweets as public service complaints or non-complaints in the Yogyakarta region.

Experience

PT. Freindcom Tech Indonesia, Bandung Barat, Indonesia *Mei 2026 – Now*
Hardware Test Engineer
- Conduct EMC and environmental testing for 1-phase and 3-phase energy meters and Data Concentration Units (DCUs).
- Develop an STM32-based mini meter simulator for DLMS communication testing

National Taiwan University of Science and Technology (NTUST), Taipei, Taiwan *Jul 2023 – Nov 2023*
Research Assistant Intern, Center for IoT Innovation (CITI)
Developed a Fault Prediction System for Wind Turbine Generators using LSTM networks in Python (TensorFlow, scikit-learn). Addressed data imbalance with SMOTEEN and implemented an Autoencoder with Attention layers, achieving MAE = 1.36 and $R^2 = 0.9$.

PT. Lunar Inovasi Teknologi, Yogyakarta, Indonesia *Dec 2022 – May 2023*
Hardware Test Engineer
Designed and developed PCBs for power monitoring ICs (ADE9153A and RN8902C) integrated with an ESP32-S3 microcontroller and OLED display to create a Smart Power Metering product using Altium Designer.

Internship Program *Sep 2022 – Nov 2022*
Designed and developed a power monitoring minimum system board utilizing RN8209C, V9021, and V9021S ICs using Altium Designer.

Project

Moneysaurus
AI-powered personal finance assistant integrated with WhatsApp and Telegram for seamless expense tracking and financial management.

- Developed OCR-based receipt parsing system for automated expense extraction.
- Implemented voice-to-text input pipeline for hands-free transaction logging.
- Built and deployed Telegram bot for real-time financial interaction.
- Designed AI agent workflow using LangGraph to orchestrate financial reasoning and tool execution.

Link: *Moneysaurus.id*

Publications

Journal of Communications and Networks (2025)(Accepted): *CSI-Based Beamforming and Localization for Massive MIMO Using Deep Learning*

IEEE Access (2025): *Multi-Task Deep Learning with Adaptive CSI Filtering for Joint Beamforming and Localization.*

IEEE Access (2024): *A Survey on Machine Learning Enhanced Integrated Sensing and Communication Systems: Architectures, Algorithms, and Applications.*

Skills & Certifications

Technical: Python, C/C++, MATLAB, CNN, LSTM, Autoencoder, Attention Mechanism, AI Agents, ESP32, PCB Design (Altium, Eagle), Microcontrollers
Certificates: DeepLearning.AI TensorFlow Developer Specialization (2021)
Languages: English, Bahasa Indonesia